

AODET: ANTI-OCCLUSION FOR ENHANCED SMALL OBJECT DETECTION IN DRONE-BASED RGBT IMAGERY

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ABSTRACT

Drone-based RGBT person detection promotes various applications such as search and rescue due to its maneuverability. While existing research predominantly concentrates on refining fusion strategies and bolstering learning mechanisms for small objects, the pervasive yet unique occlusion challenge in drone-based RGBT settings remains inadequately addressed. In this work, we address the unique challenge of occlusion in the context of RGBT small object detection, particularly emphasizing its vulnerability and the distinct characteristics it exhibits across different modalities. We propose AODet, a novel Anti-Occlusion Detector meticulously crafted to tackle the challenges posed by occlusion in drone-based RGBT object detection. Our proposed approach significantly improves the detection performance of RGBT small objects, surpassing strong baselines on two large-scale datasets, VTUAV-det and RGBTDronePerson, by 1.30 points and 2.24 points in mAP_s and mAP_{50}^{tiny} , respectively.

Index Terms— Small object detection, drone image, RGBT data, occlusion problem.

1. INTRODUCTION

Drone-based RGBT person detection [1] plays a vital role in applications like search and rescue, crowd control, and security, attributed to its wide imaging range and around-the-clock capability. The drone view lays the ground for the rapid discovery of targets in a wide range. Meanwhile, the alliance of visible (RGB) and thermal (T) modalities enables robust detection under various illumination conditions.

Current research primarily focuses either on RGBT complementary information fusion [2, 3, 4] or on enhancing the learning of small targets [5, 1, 6]. These methods effectively improve the detection accuracy of RGBT small objects. Unfortunately, a universal but also distinct challenge has not been studied thoroughly, namely the RGBT small object occlusion problem. First of all, small targets occupy fewer pixels and have a low signal-to-noise ratio, making them inherently vulnerable to occlusion. More importantly, the occlusion issue presents distinctions in RGB and thermal images.

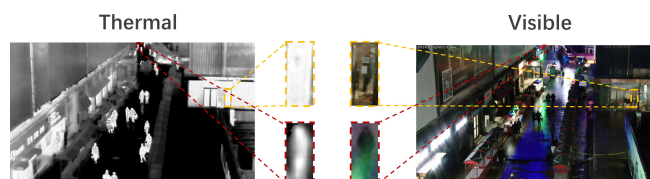


Fig. 1. Case study on the occlusion scenario. The occlusion of small targets causes significant information loss. Meanwhile, characteristics of the occlusion problem vary in RGB and thermal images.

In RGB images, occlusion stems from the blocking of visible light while it occurs in thermal images due to obstruction of thermal radiation. A straightforward example is illustrated in the yellow box of Figure 1, where the thermal radiation is obstructed by a glass wall but visible light is unimpeded. In this case, the occlusion problem only occurs in the thermal modality. Vice versa, as shown in the red box of Figure 1, the object is occluded by a flash of green light in the RGB modality but remains unspoiled in the thermal modality. The distinctions in RGBT modalities can be well exploited to study how to alleviate the impact of the occlusion problem.

In this work, we pioneer in developing an anti-occlusion detector for RGBT small objects, termed AODet. We take advantage of RGBT complementary information to achieve robustness against various occlusions. Notably, we design a Dense Occlusion Head (DOHead) to further enhance the anti-occlusion capability based on fused features. We introduce the Pyramid Shuffle technique [7] in DOHead to achieve interactions between queries in different levels so that cross-level information can be utilized. Moreover, we design a Cross-level Query Selection (CQS) to effectively select distinctive queries in densely occluded scenarios. Experiments on two large-scale drone-based RGBT detection datasets, VTUAV-det [1, 8] and RGBTDronePerson [1] show that our method achieves state-of-the-art performance, surpassing the strong baseline on mAP_s by 1.30 points on VTUAV-det and mAP_{50}^{tiny} by 2.24 points on RGBTDronePerson. Further, we validate the powerful anti-occlusion capability of AODet on a well-curated occlusion subset of VTUAV-det, on which AODet outperforms the baseline by 1.70 points on mAP_s .

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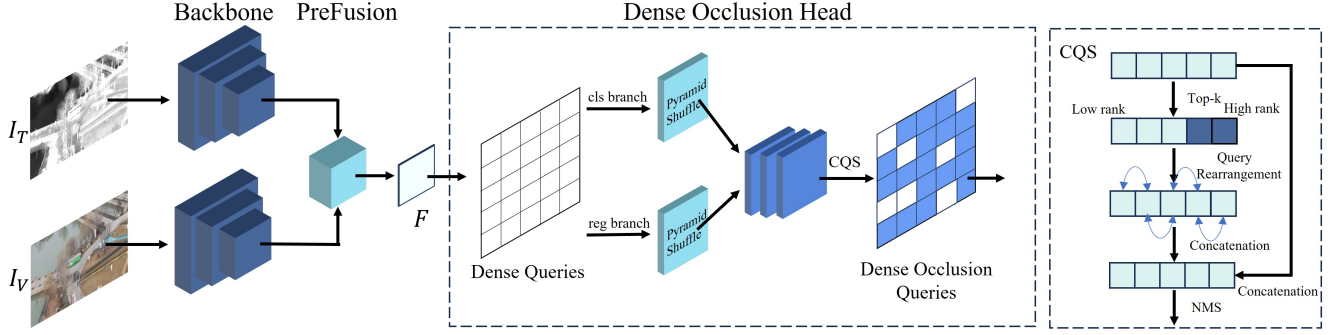


Fig. 2. Overview of AODet. Firstly the RGB and thermal features are extracted from input images and fused in the PreFusion module. Then the fused feature maps are sent to the DOHead, which is designed to enhance the detection ability of small occluded targets. Pyramid shuffle compensates for different levels of queries and CQS strengthens training on occluded small targets.

2. METHOD

2.1. Overview

The structure of our proposed AODet is shown in Figure 2. To begin with, we take a pair of RGB and thermal images as inputs and generate fused features through a PreFusion module. Since in the drone-viewed RGBT images, the illumination and thermal radiation vary in one image, the PreFusion module takes advantage of the cross-modality enhancement in Quality-aware RGBT Fusion Detector (QFDet) [1], which generates Quality-Aware Factor (QAF) maps for both modalities and adaptively fuses the multi-modal features. In this way, the RGB and thermal features can be enhanced and fused according to their quality scores across different regions. Then, the fused features are sent to the Dense Occlusion Head, where queries are densely generated from classification and regression branches. To create cross-level interactions, we adopt the Pyramid Shuffle operation on each query set. Moreover, we design a Cross-level Query Selection (CQS) to filter out redundant queries and further introduce cross-level query information in order to enhance the detection ability against dense occlusion.

2.2. Dense occlusion head

Apart from exploiting multi-modal features to improve the robustness against occlusion, we propose a Dense Occlusion Head to further alleviate the occlusion problem. The DO-Head is similar to the DDQ FCN head [7], which adopts an end-to-end detection paradigm. Notably, DOHead is different from DDQ FCN in that we design a new query selection strategy to tackle the dense occlusion problem.

As shown in Figure 2, the query is densely generated on every feature point. Then queries regarding classification and regression are yielded through each corresponding branch. To account for interactions of adjacent levels, we start by per-

forming a pyramid shuffle operation on the dense queries. For the level i feature, we interchange some channels with level $i+1$ and $i-1$ using bilinear interpolation. Afterwards, we select a number of distinct queries by CQS. These distinct queries are sent to the subsequent label assignment procedure.

Cross-level Query Selection. Previous studies have found that dense queries in end-to-end detectors achieve better recall on objects [7], while dense query end-to-end detectors are hard to optimize and fall behind conventional detectors with dynamic one-to-many label assignment strategies [10, 1]. However, in drone-viewed small object detection, there is often the case that small objects are densely distributed and occluded by each other. In this case, the NMS postprocessing in conventional detectors will suppress correctly detected dense boxes, which is unreasonable. Therefore, we adopt an end-to-end detection paradigm and propose CQS to dynamically select better queries to alleviate the negative impact of suppression.

In Figure 2, we first obtain the preliminary sorting results from the original queries through the Top-K algorithm, then select the queries of the previous and next levels, and cascade them with the queries of the current level. In this way, query information from different levels is introduced into the current level. Finally, the rearranged results are cascaded with the original query features and sent to subsequent processing. With CQS, we can not only select distinct queries from a large number of dense queries but also we consider the cross-level dependency to prevent correct queries from suppression.

2.3. Loss

Our total loss is the sum of PreHead loss L_{pre} and detection loss L_{det} , given by:

$$\begin{aligned} L_{total} &= L_{pre} + L_{det}, \\ L_{pre} &= L_{preg} + L_{pcls} + L_q, \\ L_{det} &= L_{main} + L_{aux}, \end{aligned} \quad (1)$$

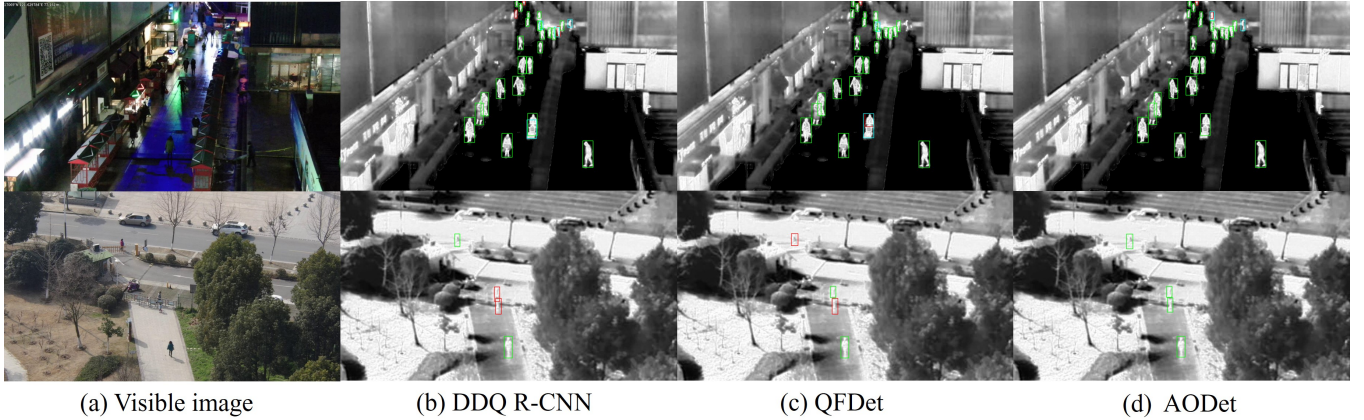


Fig. 3. Comparison of visualization results on VTUAV-det with state-of-the-art methods. Especially, the example on the second row is from the VTUAV-occlusion subset. Green boxes represent True Positives (TP), blue boxes represent False Positives (FP), and red boxes represent False Negatives (FN).

Table 1. Comparison results with state-of-the-art methods on the VTUAV-det dataset

Method	mAP	mAP ₅₀	mAP ₇₅	mAP _s	mAP _m	mAP _l	FPS
Faster R-CNN [9]	27.30	59.40	21.00	1.10	29.40	55.40	39.9
ATSS [10]	28.80	66.60	20.70	8.90	28.40	55.60	42.0
Cascade R-CNN [11]	28.20	60.80	22.10	2.50	30.20	56.10	33.5
DDQ R-CNN [7]	32.50	74.60	23.00	17.30	31.80	56.00	27.7
QFDet [1]	33.10	74.70	24.20	18.40	32.00	57.10	18.7
AODet (Ours)	34.40	77.00	25.30	19.70	33.50	57.30	18.4

where L_{pre} follows the design of QFDet PreHead [1]. The loss of the detector is divided into main loss L_{main} and auxiliary loss L_{aux} , and the classification task in the main loss is supervised by QFocal loss [12] and the regression task is supervised by GIoU loss [13]. To solve the problem of dense occlusion, inspired by DeFCN [14] and DDQ [7], we adopt an auxiliary loss that is the same as the main loss to achieve one-to-many assignment for dense occlusion queries.

3. EXPERIMENT

3.1. Datasets

We conduct experiments on two datasets: VTUAV-det [1, 8] and RGBTDronePerson [1]. We also curate a new dataset, VTUAV-occlusion, which focuses on person detection in occluded situations. We select 120 pairs of small, occluded objects from the “person” category in the VTUAV-det test set to form the VTUAV-occlusion subset.

3.2. Implementation details

All the experiments are conducted on a single NVIDIA RTX 3090 GPU, models are based on MMDetectionV2 [15]

and PyTorch [16]. Unless stated otherwise, we use ResNet-50 [17] with FPN [18] as our network backbone. The model uses the SGD optimizer for training with a learning rate of 0.01, a momentum of 0.9, and a weight decay of 0.0001. DDQ R-CNN and AODet use an AdamW optimizer with a learning rate of 0.0001 and a weight decay of 0.05. To deal with the object size imbalance problem, we reweigh the loss for different levels following [5].

3.3. Main result

Result on VTUAV-det. We convert some state-of-the-art methods [9, 10, 11] into two-stream RGBT detection networks. As shown in Table 1, mAP_s, mAP_m, mAP_l represents the average accuracy of small, medium and large targets respectively. AODet achieves the state-of-the-art performance of 34.40% mAP on the VTUAV-det dataset, surpassing QFDet by 1.3 points and DDQ R-CNN by 1.9 points. It is worth noting that our method shows an obvious advantage in detecting small targets, surpassing QFDet by 1.3 points and DDQ R-CNN by 2.4 points in terms of mAP_s. At the same time, our method is also highly competitive for medium and large targets. Shown in Table 3, our experiments on the VTUAV-occlusion dataset demonstrate that AODet surpasses QFDet by 1.7 points and DDQ R-CNN by 2.9 points, which

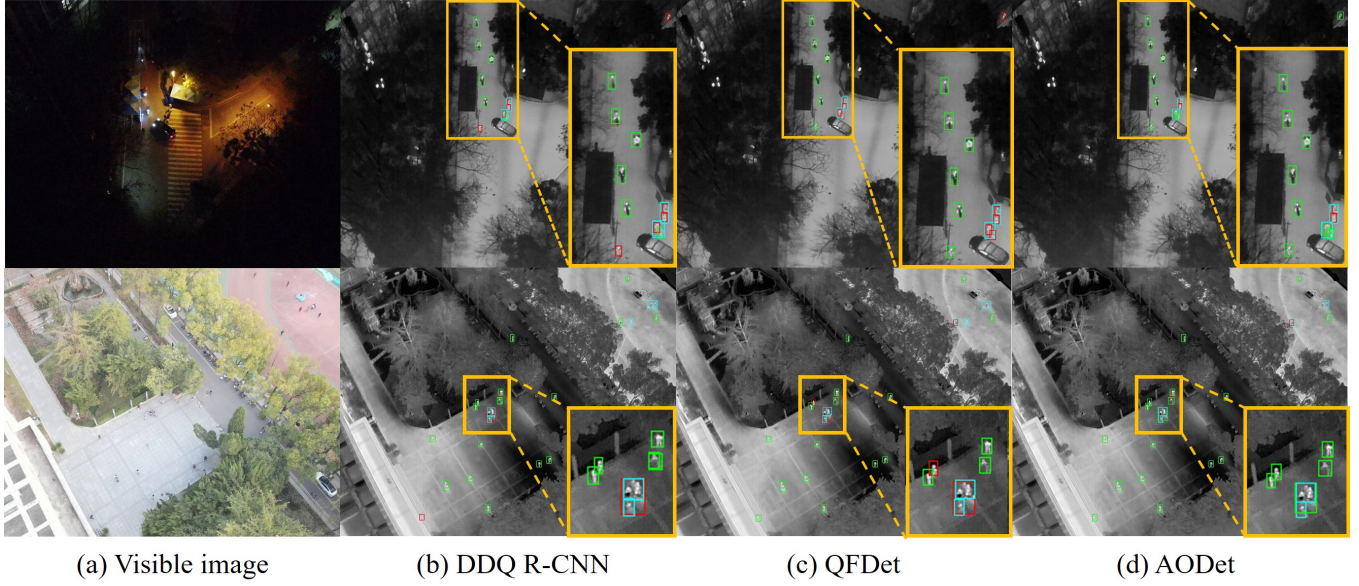


Fig. 4. Comparison of visualization results on RGBTDronePerson with state-of-the-art methods. Green boxes represent True Positives (TP), blue boxes represent False Positives (FP), and red boxes represent False Negatives (FN).

Table 2. Comparison with state-of-the-art methods on the RGBTDronePerson dataset

Method	mAP_{50}^{all}	mAP_{50}^{tiny}	mAP_{50}^{tiny1}	mAP_{50}^{tiny2}	mAP_{50}^{tiny3}	mAP_{50}^{small}	mAP_{25}^{all}
Faster R-CNN [9]	26.70	27.09	0.00	21.03	32.99	32.15	36.86
ATSS [10]	36.16	37.20	14.19	23.00	44.07	27.73	52.91
Cascade R-CNN [11]	29.07	29.82	0.00	26.09	34.96	28.31	40.08
DDQ R-CNN [7]	45.92	47.16	22.53	36.83	53.68	30.64	59.30
QFDet [1]	44.70	46.65	22.12	34.58	53.07	27.22	61.13
AODet (Ours)	47.66	49.40	22.66	37.14	56.43	29.39	60.99

strongly proves the effectiveness of our method in solving the RGBT small object occlusion problem.

Table 3. Comparison with state-of-the-art methods on the VTUAV-occlusion dataset

Method	mAP	mAP_s	mAP_m	mAP_l
Faster R-CNN [9]	26.80	1.00	27.50	45.60
ATSS [10]	27.00	15.00	24.80	48.00
Cascade R-CNN [11]	27.40	1.30	28.00	47.50
DDQ R-CNN [7]	32.40	29.00	30.00	48.90
QFDet [1]	32.70	30.20	30.20	49.20
AODet (Ours)	33.20	31.90	30.50	50.00

Result on RGBTDronePerson. To better verify the generalizability of our method, we also conducted comparative experiments with the state-of-the-art methods on the RGBTDronePerson dataset. Since the target size in the RGBTDronePerson dataset is smaller, we further refine the small targets into tiny and small. In Table 2, “tiny” and

“small” represent absolute size are in the range of [0,20) and [20,32), AODet achieves the mAP_{50}^{all} of 47.66% and the mAP_{50}^{tiny} of 49.40%, outperforming the second method by 1.74 points and 2.24 points.

4. CONCLUSION

In this study, we propose an anti-occlusion detector for drone-based RGBT person detection, addressing the issue of occlusion. To enhance small target detection, particularly in scenarios involving occluded scenarios, we devise the Cross-level Query Selection. Extensive experiments are conducted on two large RGBT person detection datasets, namely RGBTDronePerson and VTUAV-det, along with our specifically constructed occlusion subset, VTUAV-occlusion. The compelling results showcase that our proposed method consistently demonstrates superior performance across diverse scenarios.

5. REFERENCES

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